A counterexample to a weighted-geometric-mean log-majorization conjecture

Candidate counterexample for arXiv:2105.13356

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Status. Candidate counterexample, likely valid, requiring human review. Exact integer positive-definite matrices disprove Conjecture 1.1 of Ghabries–Abbas–Mourad–Assi. The decisive numerical inequality is certified by exact rational ball arithmetic.

1 Source conjecture

For positive-definite matrices A, B, their weighted geometric mean is

$$A \,\sharp_t\, B = A^{1/2}(A^{-1/2}BA^{-1/2})^t A^{1/2}, \qquad 0 \le t \le 1.$$

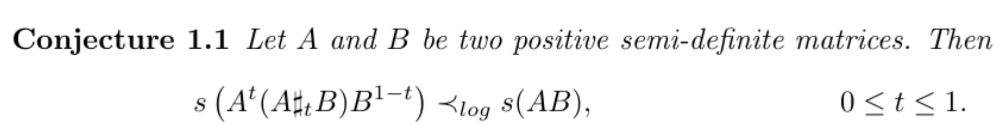
Writing $s_1(Y) \ge \cdots \ge s_n(Y)$ for the singular values of Y, one says $s(X) \prec_{\log} s(Y)$ when

$$\prod_{j=1}^{k} s_j(X) \le \prod_{j=1}^{k} s_j(Y) \quad (1 \le k < n), \qquad |\det X| = |\det Y|.$$

Conjecture 1.1 in Ghabries–Abbas–Mourad–Assi [1] asks whether

$$s\big(A^t(A \,\sharp_t\, B)B^{1-t}\big) \prec_{\log} s(AB) \qquad (0 \le t \le 1) \tag{1}$$

for all positive semidefinite A, B. It originates in work of Lemos and Soares [2]. The source statement is reproduced below.



Conjecture 1.1 *Let A and B be two positive semi-definite matrices. Then*

$$s\left(A^t(A\sharp_t B)B^{1-t}\right) \prec_{log} s(AB), \qquad 0 \le t \le 1.$$

Figure 1: Conjecture 1.1 in the source PDF [1, p. 3].

2 Idea of the proof

The determinant in (1) always agrees on the two sides. Consequently, in dimension four its third prefix-product inequality says exactly that the smallest singular value of the left-hand matrix is at least the smallest singular value of AB. This dual endpoint is where the conjecture fails.

The example below makes one matrix diagonal and uses a rational value $t = 15/16$, so all fractional powers can be built from four successive positive square roots. Exact LDL^{T} arithmetic proves a lower bound for $s_{\min}(AB)$. Repeated rational matrix balls enclose the square roots and give an upper bound for the other smallest singular value. The two certified bounds are separated by more than 0.08.

3 The counterexample

Theorem 1. *Let $t = 15/16$ and let*

$$A = \begin{pmatrix} 17825066 & 0 & 0 & 0 \\ 0 & 83 & 0 & 0 \\ 0 & 0 & 24 & 0 \\ 0 & 0 & 0 & 10 \end{pmatrix}, \qquad B = \begin{pmatrix} 117567 & -99513 & 2310 & -38624 \\ -99513 & 131686 & 572 & 24611 \\ 2310 & 572 & 190 & -1191 \\ -38624 & 24611 & -1191 & 14107 \end{pmatrix}.$$

Then A and B are positive definite, but

$$s\big(A^t(A \natural_t B)B^{1-t}\big) \not\prec_{\log} s(AB).$$

More precisely, if $X = A^t(A \natural_t B)B^{1-t}$, then

$$s_{\min}(X) < 235.715744 < 235.8 < s_{\min}(AB). \tag{2}$$

Numerically, the violated third-prefix ratio is

$$\frac{\prod_{j=1}^{3} s_j(X)}{\prod_{j=1}^{3} s_j(AB)} = 1.0009342385069400\ldots > 1.$$

Proof. The positivity of A is immediate. The four leading principal minors of B are

$$117567, \quad 5579090793, \quad 55894510422, \quad 2308323855975,$$

so $B > 0$ by Sylvester's criterion.

For arbitrary positive-definite A, B,

$$\det(A \natural_t B) = (\det A)^{1-t}(\det B)^t.$$

It follows that

$$\det X = (\det A)^t \det(A \natural_t B)(\det B)^{1-t} = \det A \det B = \det(AB). \tag{3}$$

Because all singular values are positive, (3) gives

$$\frac{\prod_{j=1}^{3} s_j(X)}{\prod_{j=1}^{3} s_j(AB)} = \frac{s_{\min}(AB)}{s_{\min}(X)}.$$

Thus it remains only to prove (2).

Set $q = 1179/5 = 235.8$. Exact integer multiplication gives that

$$D = 25(AB)^{\mathsf{T}}(AB) - 1179^2 I_4$$

has leading principal minors

$$\begin{aligned} \Delta_1 &= 109792626457520745575173084, \\ \Delta_2 &= 42609956693745032985833310177535659834256, \\ \Delta_3 &= 2228978970825693898612895858077918124109611904, \\ \Delta_4 &= 182721114621736343069686961423087938455547419076911. \end{aligned}$$

All are positive. Hence $D > 0$ and therefore $(AB)^{\mathsf{T}}(AB) > q^2 I_4$, proving $s_{\min}(AB) > q$.

We now certify the other half of (2). The included verifier uses operator-norm balls with exact rational centers and radii. Its square-root step rests on the following elementary bound. If $S, R > 0$, then

$$\|R - S^{1/2}\|_F \le \frac{\|R^2 - S\|_F}{\lambda_{\min}(R) + \sqrt{\lambda_{\min}(S)}}. \tag{4}$$

Indeed, for $E = R - S^{1/2}$ one has $RE + ES^{1/2} = R^2 - S$; the Sylvester operator on the left has least singular value at least $\lambda_{\min}(R) + \sqrt{\lambda_{\min}(S)}$ in the Frobenius norm.

Starting from the exact matrices A, B, four applications of (4) enclose the $1/16$ powers of A, B, and $A^{-1/2}BA^{-1/2}$. Products and inverses use the standard norm-ball rules

$$e_{UV} \le \|U_0\|e_V + \|V_0\|e_U + e_U e_V, \qquad e_{U^{-1}} \le \frac{\|U_0^{-1}\|^2 e_U}{1 - \|U_0^{-1}\|e_U}.$$

All displayed norms are bounded above by exact rational Frobenius bounds, and all eigenvalue lower bounds use exact rational inverses and LDL^{T} pivots. The resulting ball for X has radius less than 2.546×10^{-42}. An exact rational trial vector v then gives

$$s_{\min}(X) \le \frac{\|Xv\|}{\|v\|} < 235.715744.$$

Every comparison in this last paragraph is executed as an exact rational inequality by `code/certify_counterexample.py`; floating point is used only to select the rational centers and trial vector. This proves (2), hence the third prefix inequality is reversed and (1) is false. □

4 Verification and scope

Running

```
conda run --no-capture-output -n sandbox python code/certify_counterexample.py
```

reproduces the rational certificate. Its high-precision centers give

$$s_{\min}(X) = 235.715743188457775258697\ldots, \quad s_{\min}(AB) = 235.935957912436422532775\ldots.$$

The separate search script records the heuristic discovery route; it is not used in the proof.

The counterexample is strict, so continuity yields an open interval of failing parameters around 15/16. No claim is made that dimension four is minimal, and the source's other conjectures are not decided here.

Bounded novelty check. The run's registry, solution, attempt, and proof-gap indexes; exact expressions; the source title, authors, and DOI; weighted geometric-mean and log-majorization keywords; and the source's citing works were searched through 11 August 2026. Gan and Kim reproduce the exact conjecture and call it still open [3, Sec. 3]. The three indexed citing papers through that date do not state a decision of this conjecture, and no prior counterexample was found. This is a bounded audit, not a definitive priority claim.

Human-review recommendation. First audit the exact matrix transcription and the determinant reduction (3). Then inspect the square-root enclosure (4), the propagation of ball radii, and the final exact rational assertions in the verifier. These are the only computer-assisted parts of the proof.

References

[1] M. M. Ghabries, H. Abbas, B. Mourad, and A. Assi, *New log-majorization results concerning eigenvalues and singular values and a complement of a norm inequality*, Linear Multilinear Algebra 71 (2023), 1228–1243; arXiv:2105.13356.

[2] R. Lemos and G. Soares, *Some log-majorizations and an extension of a determinantal inequality*, Linear Algebra Appl. 547 (2018), 19–31.

[3] L. Gan and S. Kim, *Weak log-majorization between the geometric and Wasserstein means*, J. Math. Anal. Appl. 530 (2024), 127711; arXiv:2301.07934.